

On the Impact of Face Segmentation-Based Background Removal on Recognition and Morphing Attack Detection

Eduarda Caldeira^{1,2}, Guray Ozgur^{1,2}, Fadi Boutros¹, Naser Damer^{1,2}

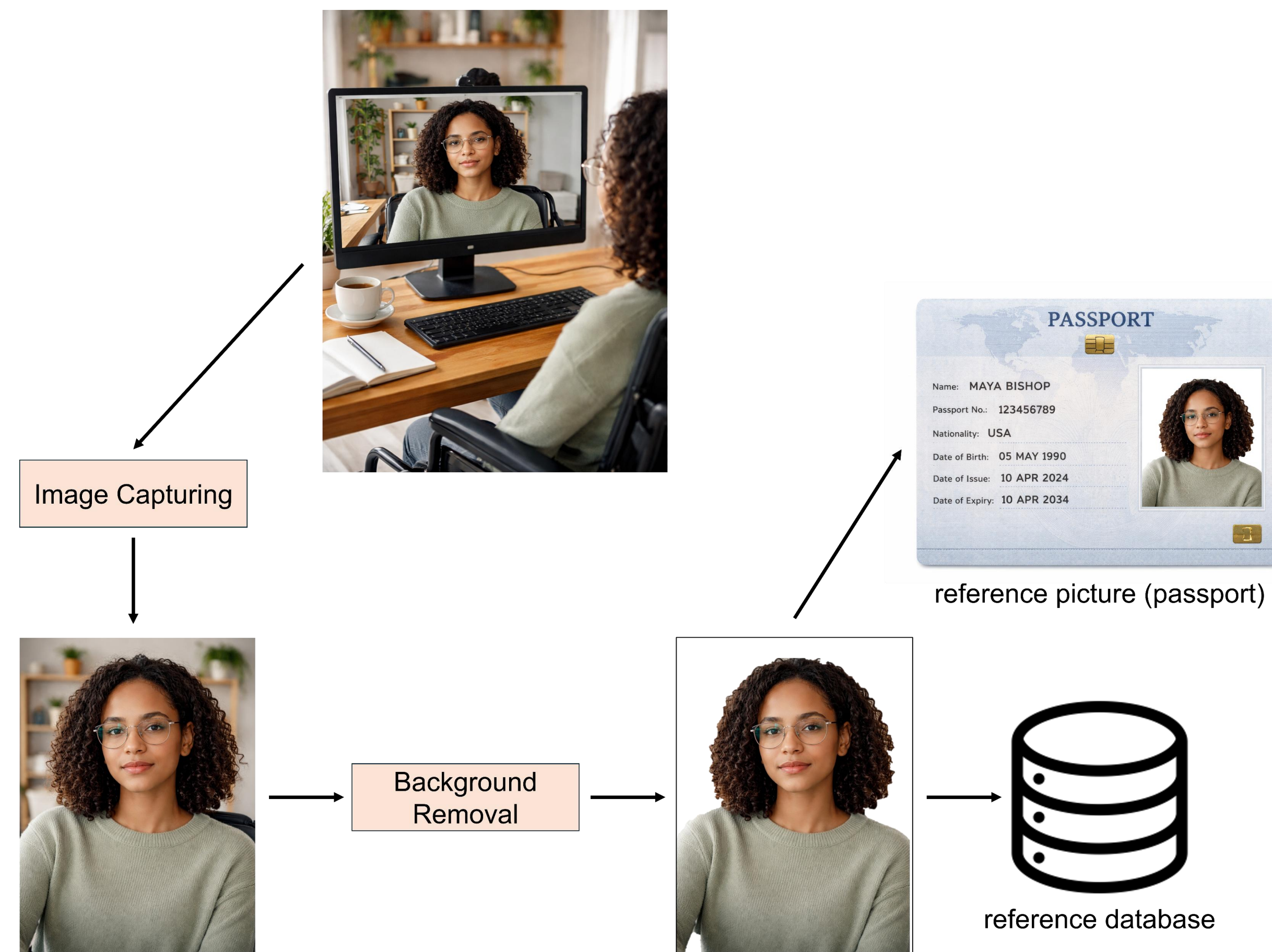
¹ Fraunhofer Institute for Computer Graphics Research IGD, Germany

² Department of Computer Science, TU Darmstadt, Germany

Problem Statement

Biometric systems deployment in large-scale identity management systems, such as the European Entry/Exit System, has intensified the need for reliable face image acquisition under unconstrained conditions. While large-scale biometric systems require high-quality enrollment images that adhere to specific constraints (such as the presence of a plain, light-colored background), these requirements are hard to satisfy in real-world scenarios.

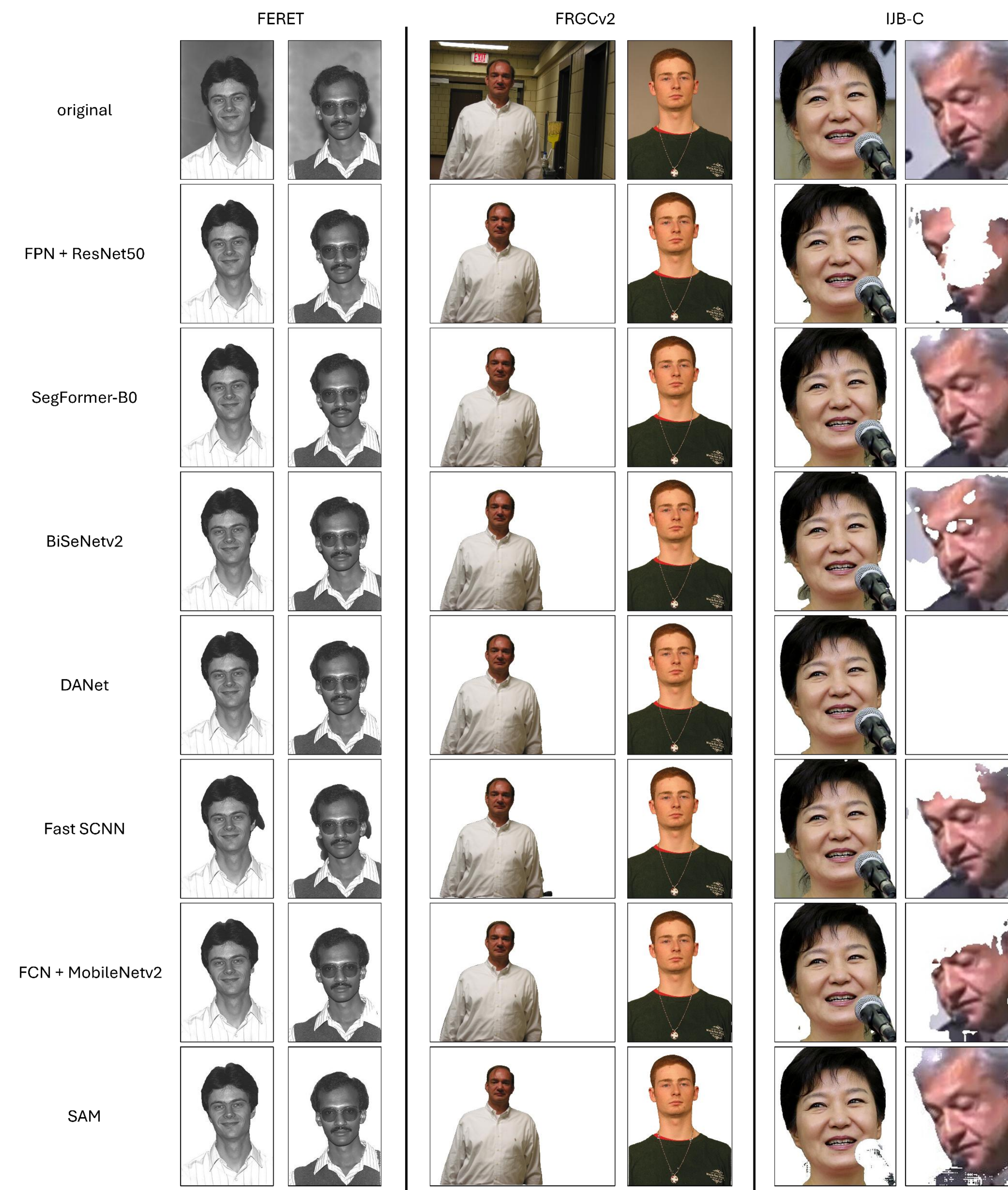
Despite the appeal of face segmentation and background correction solutions to standardize image captures, these preprocessing steps fundamentally modify their original content, making it essential to assess the extent to which they affect the reliability and security of biometric systems.



Research Questions

We investigate the impact of face segmentation-based background removal on the biometric processing pipeline, addressing two research questions:

1. How do different background removal strategies affect **face image quality** and the performance in downstream tasks such as **face recognition** and **morphing attack detection**?
2. How do the **image acquisition conditions** affect downstream tasks' performance when background removal is performed?



Conclusions

1. Background removal impacts **FR performance** for images captured in uncontrolled conditions
2. The **false triggering of MAD systems** is highly method-dependent, with foundation-model-based methods suffering significant degradation after background removal
3. Hence, **background removal is not a neutral operation** and should be treated as an integral component of the biometric pipeline

Face Recognition

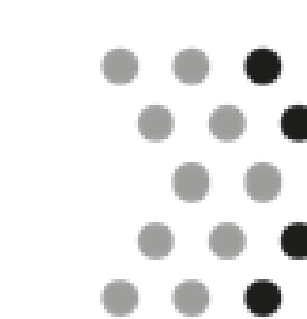
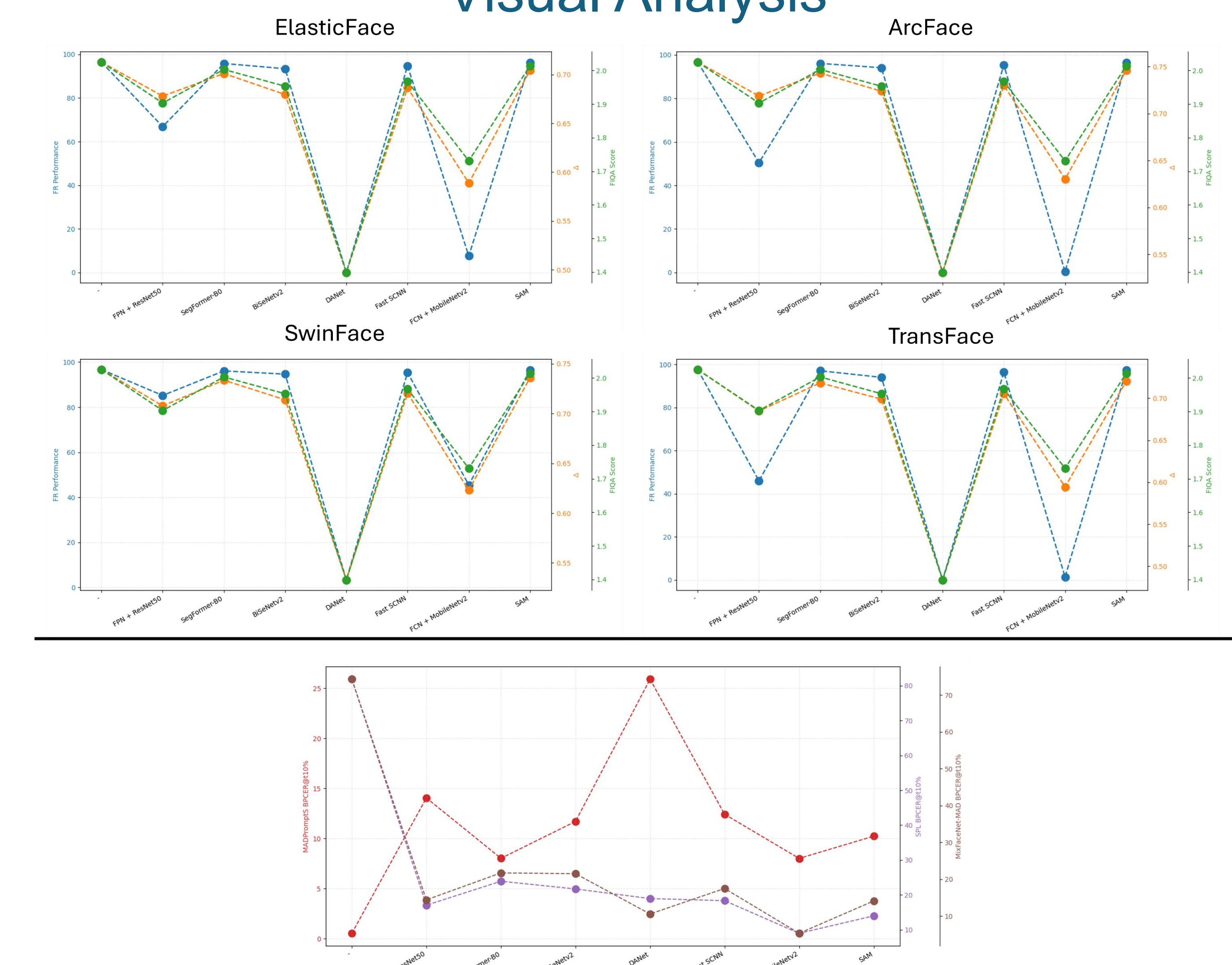
Dataset	Segmentation Method	FR Performance (%)				FIQA	
		ElasticFace	ArcFace	SwinFace	TransFace	avg	std
FERET	—	100.00	100.00	100.00	100.00	2.370	0.074
	FPN + ResNet50 [30]	100.00	100.00	100.00	100.00	2.369	0.074
	SegFormer-B0 [53]	100.00	100.00	100.00	100.00	2.369	0.074
	BiSeNetv2 [55]	100.00	100.00	100.00	100.00	2.367	0.075
	DANet [19]	100.00	100.00	100.00	100.00	2.369	0.074
	Fast SCNN [42]	100.00	100.00	100.00	100.00	2.368	0.074
FRGCv2	FCN + MobileNetv2 [32]	100.00	100.00	100.00	100.00	2.368	0.074
	SAM [27]	100.00	100.00	100.00	100.00	2.362	0.076
	—	99.98	99.98	99.98	99.98	2.329	0.071
	FPN + ResNet50 [30]	99.99	99.99	99.99	99.99	2.329	0.070
	SegFormer-B0 [53]	99.99	99.99	99.99	100.00	2.329	0.070
	BiSeNetv2 [55]	99.99	99.99	99.99	99.99	2.329	0.071

Segmentation Method	FR Performance (%)				FIQA	
	ElasticFace	ArcFace	SwinFace	TransFace	avg	std
—	94.67	96.47	95.25	96.71	96.38	2.025
FPN + ResNet50 [30]	94.67	96.47	95.25	96.71	96.38	2.025
SegFormer-B0 [53]	94.67	96.47	95.25	96.71	96.38	2.025
BiSeNetv2 [55]	94.67	96.47	95.25	96.71	96.38	2.025
DANet [19]	94.67	96.47	95.25	96.71	96.38	2.025
Fast SCNN [42]	94.67	96.47	95.25	96.71	96.38	2.025
FCN + MobileNetv2 [32]	94.67	96.47	95.25	96.71	96.38	2.025
SAM [27]	94.67	96.47	95.25	96.71	96.38	2.025

Morphing Attack Detection

Dataset	Segmentation Method	MAD Performance (%)				MAD PromptS		
		BPCER@1%	SPL	MixFaceNet-MAD	BPCER@10%	BPCER@1%	BPCER@10%	BPCER@20%
FERET	—	45.89	25.13	18.07	100.00	36.22	4.81	1.18
	FPN + ResNet50 [30]	28.50	16.92	13.15	100.00	6.54	8.61	2.57
	SegFormer-B0 [53]	28.91	16.79	13.30	100.00	6.25	8.34	2.50
	BiSeNetv2 [55]	25.72	15.69	12.71	100.00	6.76	8.41	2.28
	DANet [19]	25.09	15.94	13.04	100.00	6.58	8.42	2.39
	Fast SCNN [42]	26.49	15.47	12.49	100.00	6.32	8.38	2.17
FRGCv2	FCN + MobileNetv2 [32]	25.20	15.28	12.09	100.00	5.69	8.01	2.79
	SAM [27]	18.81	9.70	7.83	100.00	7.27	62.05	12.27
	—	65.19	45.87	38.22	100.00	49.80	31.96	1.21
	FPN + ResNet50 [30]	28.52	17.74	13.47	100.00	6.25	2.81	2.94
	SegFormer-B0 [53]	27.34	16.56	12.62	100.00	5.51	2.34	3.10
	BiSeNetv2 [55]	27.57	16.20	12.00	100.00	5.79	2.71	2.76
IJB-C	DANet [19]	27.70	16.32	12.24	100.00	6.13	2.82	2.49
	Fast SCNN [42]	27.20	16.17	12.12	100.00	6.01	2.77	2.88
	FCN + MobileNetv2 [32]	27.07	15.69	11.65	100.00	5.71	2.64	2.85
	SAM [27]	18.03	10.92	8.03	100.00	3.33	1.34	11.89
	—	86.98	81.83	78.73	100.00	74.29	63.03	0.17
	FPN + ResNet50 [30]	24.62	17.07	14.13	100.00	14.39	5.97	7.13

Visual Analysis



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